

Machine Learning

Assignment 1

Question 1: Decision Tree Construction (AND Function) [CLO- 1]

Part1) Full dataset

x1	x2	x3	y
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

Part2) Determine the best feature for the root node

We usually choose that feature that gives the best split at the root and the best split in a decision tree is the one with the highest Information Gain.

1) Finding the entropy of the whole dataset

For the 3-variable AND function, the dataset has:

- 7 negative case in which (output 0)
- 1 positive case in which (output 1)

So the probability of each class is:

$$p(1) = \frac{1}{8}, \quad p(0) = \frac{7}{8}$$

Formula of Entropy

$$H(S) = -\sum p_i \log_2(p_i)$$

$$H(S) = -p(1)\log_2 p(1) - p(0)\log_2 p(0)$$

Substitute values:

$$H(S) = -\left(\frac{1}{8}\right)\log_2\left(\frac{1}{8}\right) - \left(\frac{7}{8}\right)\log_2\left(\frac{7}{8}\right)$$
$$H(S) \approx 0.54$$

$H(S) \approx 0.54$ this is the uncertainty in the dataset without splitting.

2) Split the dataset using x_1

Now divide the dataset into two groups based on x_1 .

Lets take $x_1 = 0$

x1	x2	x3	y
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0

So all outputs are 0. That means this group is pure, meaning

$$H(S_{x_1=0}) = 0$$

Lets take $x_1 = 1$

1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

So here we have:

- 3 zeros
- 1 one

Probabilities:

$$p(0) = \frac{3}{4}, p(1) = \frac{1}{4}$$

Entropy of this group:

$$H(S_{x_1=1}) = -\left(\frac{1}{4}\right) \log_2 \left(\frac{1}{4}\right) - \left(\frac{3}{4}\right) \log_2 \left(\frac{3}{4}\right)$$

$$H(S_{x_1=1}) \approx 0.811$$

3) Computing the expected entropy after splitting on x_1

Now combine the entropies of both groups, weighted by how many samples are in each group.

Each group has 4 samples out of 8, so:

$$\begin{aligned} \text{Expected Entropy}(x_1) &= \left(\frac{4}{8}\right)(0) + \left(\frac{4}{8}\right)(0.811) \\ &= 0 + 0.4055 \end{aligned}$$

$$\text{Expected Entropy}(x_1) \approx 0.405$$

Step 4: Compute Information Gain for x_1

Information Gain is:

$$\begin{aligned} \text{IG}(x_1) &= H(S) - \text{Expected Entropy}(x_1) \\ \text{IG}(x_1) &= 0.543 - 0.405 \\ \text{IG}(x_1) &= 0.137 \end{aligned}$$

So splitting on x_1 reduces uncertainty by 0.137.

Now Checking for x_2 and x_3

For the AND function, all three features behave the same way because of AND.

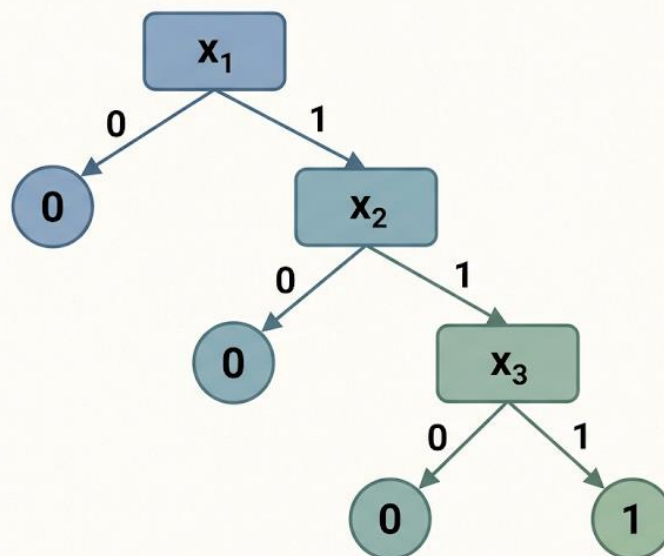
so

$$IG(x_1) = IG(x_2) = IG(x_3)$$

5) Choosing the root node

The best root feature is any of x_1 , x_2 , or x_3 because they all give the same Information Gain, and we can choose x_1 as the root node.

Part3) Draw the final tree clearly.



Question 2: Gradient Descent on a Quadratic Loss [CLO- 1]

Given the loss function:

$$L(w) = (w - 3)^2 + 2$$

Initial value: $w_0 = 0$

Learning rate: $\eta = 0.1$

1. Gradient of the Loss Function

To compute the gradient, we differentiate the loss function with respect to w :

$$\frac{dL}{dw} = 2(w - 3)$$

2. Perform Two Iterations of Gradient Descent

The update rule is:

$$w_{\text{new}} = w_{\text{old}} - \eta \cdot \frac{dL}{dw}$$

First Iteration:

At $w_0 = 0$:

$$\begin{aligned}\frac{dL}{dw}(0) &= 2(0 - 3) = -6 \\ w_1 &= 0 - 0.1(-6) = 0.6\end{aligned}$$

Second Iteration:

At $w_1 = 0.6$:

$$\begin{aligned}\frac{dL}{dw}(0.6) &= 2(0.6 - 3) = -4.8 \\ w_2 &= 0.6 - 0.1(-4.8) = 1.08\end{aligned}$$

3. Compute Loss at Each Step

$$\begin{aligned}L(w_0) &= (0 - 3)^2 + 2 = 11 \\ L(w_1) &= (0.6 - 3)^2 + 2 = 7.76 \\ L(w_2) &= (1.08 - 3)^2 + 2 = 5.6864\end{aligned}$$

Question 3: K-Means Clustering (Unsupervised Learning) [CLO-2]

Given the dataset:

$$X = \{(1,1), (2,1), (4,3), (5,4)\}$$

and the initial centroids:

$$\mu_1 = (1,1) \quad \mu_2 = (5,4)$$

*****First time Running*****

1. Assign each point to the nearest centroid

We use Euclidean distance to find the nearest centroid.

For point (1, 1):

- Distance to $\mu_1 = (1,1)$

$$\sqrt{(1-1)^2 + (1-1)^2} = 0$$

- Distance to $\mu_2 = (5,4)$

$$\sqrt{(5-1)^2 + (4-1)^2} = 5$$

So, (1, 1) is assigned to **Cluster 1**.

For point (2, 1):

- Distance to $\mu_1 = (1,1)$

$$\sqrt{(1-2)^2 + (1-1)^2} = 1$$

- Distance to $\mu_2 = (5,4)$

$$\sqrt{(5-2)^2 + (4-1)^2} = \sqrt{18} \approx 4.24$$

So, (2, 1) is assigned to **Cluster 1**.

For point (4, 3):

- Distance to $\mu_1 = (1,1)$

$$\sqrt{(4-1)^2 + (3-1)^2} = \sqrt{13} \approx 3.61$$

- Distance to $\mu_2 = (5,4)$

$$\sqrt{(5-4)^2 + (4-3)^2} = \sqrt{2} \approx 1.41$$

So, (4, 3) is assigned to **Cluster 2**.

For point (5, 4):

- Distance to $\mu_1 = (1,1)$ is 5
- Distance to $\mu_2 = (5,4)$ is 0

So, (5, 4) is assigned to **Cluster 2**.

2. Computing new centroids

Now we find the mean of the points in each cluster.

For **Cluster 1**:

$$\begin{aligned}\mu_1 &= \left(\frac{1+2}{2}, \frac{1+1}{2} \right) \\ &= (1.5, 1)\end{aligned}$$

For **Cluster 2**:

$$\begin{aligned}\mu_2 &= \left(\frac{4+5}{2}, \frac{3+4}{2} \right) \\ &= (4.5, 3.5)\end{aligned}$$

Second time Running

3. Performing one new reassignment of centroids

Now we will check if any point changes cluster using the new centroids.

For point (1, 1):

- It is closer to (1.5, 1), so it stays in **Cluster 1**

For point (2, 1):

- It is also closer to (1.5, 1), so it stays in **Cluster 1**

For point (4, 3):

- It is closer to (4.5, 3.5), so it stays in **Cluster 2**

For point (5, 4):

- It is also closer to (4.5, 3.5), so it stays in **Cluster 2**

Since the cluster assignments do not change, the centroids remain the same:

$$\mu_1 = (1.5, 1) \quad , \quad \mu_2 = (4.5, 3.5)$$

Final cluster assignments

Cluster 1:

$$\{(1, 1), (2, 1)\}$$

Cluster 2:

$$\{(4, 3), (5, 4)\}$$

Question 4: Soft Margin SVM and Slack Variables [Ungraded]

1. Role of slack variables ξ_i

Basically the main purpose for slack variables is they show how much a data point breaks the margin rule.

They make SVM more flexible by allowing some points to be a little wrong instead of forcing every point to fit perfectly.

2. How slack variables allow violations of the margin constraint

In hard-margin SVM, every point must satisfy:

$$y_i(w^T x_i + b) \geq 1$$

In soft-margin SVM, the rule becomes:

$$y_i(w^T x_i + b) \geq 1 - \xi_i$$

This means a point can:

- stay outside the margin if $\xi_i = 0$
- go inside the margin if $0 < \xi_i < 1$
- be misclassified if $\xi_i > 1$

So, slack variables allow small mistakes.

3. How this helps with non-separable and noisy data

Non-separable data:

Sometimes the data cannot be split perfectly with one straight line. Hard-margin SVM may fail in that case. Soft-margin SVM works better because it allows some points to violate the rule.

Noisy data:

Real data often has outliers or wrong points. If SVM tries to fit every point exactly, the boundary may become too strict. Slack variables help the model ignore a few bad points and still find a better boundary.

4. Role of parameter C

Parameter C controls how much we punish these violations

.

- **Large C :** in this case mistakes are punished more, so the model tries to classify almost all points correctly (more acting like hard margin)
- **Small C :** in this case mistakes are allowed more easily, so the model gives a wider margin. And is less strict

So, basically C controls the balance between a strict boundary and a flexible boundary.